

MERIDIAN FAMILY OFFICE

AI Copilot — Test Suite & Results

Verifying the deterministic engine, the deliverable layer, and the end-to-end app

65/65 tests passed · 10.83s · 15 July 2026

Runs fully offline in DEMO_MODE — no API key, no network, no billing.

Coverage at a glance

Central invariant: every number is deterministic; documents and instructions shape prose, never figures.

Engine

3 files · 30 tests

Parsing · reconciliation · suitability · grounded Q&A

Deliverable

3 files · 21 tests

Doc ingestion · portable prompt · narrative · deck render

App

1 file · 14 tests

Streamlit end-to-end across every view

What's proven end-to-end

- The v10 fix: research/other documents and tactical instructions change the deterministic commentary and every exported deck (HTML, PPTX, PDF).
- The same book reads differently by mandate: advisory flags, discretionary hard-blocks.
- Reconciliation catches the ~\$436 custodian break instead of trusting it silently.

Statement ingestion

9/9 PASS

Engine · tests/test_statement_parser.py

Parse the three custodian formats, reconcile against each statement's own total, convert FX to USD, and keep a PII-free continuity key. Every downstream number rides here.

- ✓ Parse custodian A (USD, per-share CSV) into valued positions.
- ✓ Semicolon CSV, European decimals, EUR/CHF → converted to a USD base.
- ✓ JSON custodian carries illiquid alternatives + a Lombard loan (a liability).
- ✓ One sample deliberately fails its checksum by ~\$436 — it must be flagged, not silently trusted.
- ✓ The ~\$436 break on custodian B reconciles under a generous tolerance but fails under a tight one — proving the checksum knob actually gates the flag.
- ✓ Continuity key is deterministic across re-parses and carries no PII.
- ✓ Continuity check compares month-to-month keys.
- ✓ Consolidate all three custodians into one holdings list.
- ✓ gross – debt = net, and gross is the sum of position base values.

The deterministic policy checks. The mandate decides what a breach MEANS; plus concentration, liquidity, unhedged-FX, hard regulatory blocks and instruction-diff.

- | | |
|---|---|
| ✓ A single overweight equity book, evaluated under each mandate, yields the mandate-appropriate enforcement on the band breach. | ✓ Sanctioned instrument hard blocks even execution only. |
| ✓ A balanced book inside every band produces no actionable breach. | ✓ Complex product appropriateness blocks when not allowed. |
| ✓ A single stock above the 10% single-security cap is caught; a diversified fund of the same weight is not. | ✓ Client asked for a treasury sleeve; the proposal holds a corporate-credit bond fund of the wrong duration → a deviation flag. |
| ✓ Parked liquidity (money-market) is exempt from the single-position cap. | ✓ Instruction diff satisfied when matched. |
| ✓ Below the min-liquid floor (here mostly illiquid alternatives) is flagged. | ✓ Worst enforcement ranking. |
| ✓ Economic FX exposure above the max-unhedged cap is flagged. | ✓ The three-custodian sample book, under its advisory profile, lands on a 'flag' gate (bands breached, nothing hard-blocked). |

Routing picks the right deterministic tool from the phrasing; answers stay grounded in computed facts; the router never crashes on odd input.



'Net worth' routes to the totals tool.



An FX / currency question routes to the currency tool.



A suitability / mandate question routes to the suitability tool.



A 'largest position' question routes to the concentration tool.



A liquidity question routes to the liquidity tool.



A reconciliation question routes to the data-quality tool.



Every answer carries facts tied to the computed book.



The deterministic briefing fact-sheet spans every domain.



The router must always resolve to SOME tool — never crash on odd input.

The research / other uploaders read text deterministically so content can be folded into the prompt as advisory context — never a source of figures.

- ✓ Plain text and Markdown read cleanly.
- ✓ HTML is stripped to readable text (tags removed).
- ✓ An .eml email body is extracted.
- ✓ CSV and JSON pass through as text.
- ✓ An unsupported file type fails gracefully (no crash).
- ✓ Very large documents are truncated (MAX_CHARS) and marked, never dropped.
- ✓ The extraction result always has the same shape.

The portable prompt bundles every input; the keyless commentary REFLECTS the documents (the v10 fix) while only ever quoting the computed numbers.

- ✓ The prompt carries every block (FACTS, intake, holdings, docs, tactical).
- ✓ Document text is embedded verbatim in the prompt.
- ✓ Without documents, that content is absent from the prompt.
- ✓ The FACTS block is valid JSON and holds the numbers.
- ✓ v10 fix: documents change the deterministic commentary.
- ✓ The commentary quotes short excerpts of the tactical text and docs.
- ✓ The keyless summary must use the model's own numbers verbatim and never invent a figure that isn't already somewhere in the computed model.
- ✓ The commentary carries the 'not investment advice' disclaimer.

HTML / PPTX / PDF renderers agree, figures are copied from the model, and documents change every exported format.

- ✓ All three renderers (HTML/PPTX/PDF) produce valid output.
- ✓ The deck shows the computed net-worth & gross figures.
- ✓ The commentary slide appears only when a narrative is set.
- ✓ v10 fix: documents change the HTML, PPTX and PDF exports.
- ✓ With documents, the deck surfaces the client's liquidity event.
- ✓ The deck carries the 'not independently verified' disclaimer.

Drive the real Streamlit app across the views a banker walks through: load samples, review each view, change the mandate & tactical instructions — and never error.

- ✓ The app boots with no book loaded.
- ✓ Each sample statement loads into the Overview view. (3×)
- ✓ Overview / Holdings / Suitability / Proposal all render. (4×)
- ✓ The Proposal view shows the assembled prompt and commentary.
- ✓ Tactical instructions flow into the on-screen prompt.
- ✓ Under discretionary, band breaches HARD-BLOCK; under advisory they only flag. The same book therefore reads differently on the Suitability view. (2×)
- ✓ A single-custodian book still produces a proposal.
- ✓ The Proposal view with no book degrades gracefully.

How to run

python3 -m pytest

python3 -m pytest tests/test_suitability.py -v

65/65 passed in 10.83s · offline, deterministic, repeatable